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A hierarchical attention network integrating multi-scale relationship for drug response prediction

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ABSTRACT

Anticancer drug response prediction with deep learning technology has become the foundation of precision medicine. It is essential for anticancer drug response prediction to incorporate multi-scale relationships within feature items and biomedical entities. Therefore, we propose MultiDRP that develops the hierarchical attention networks integrating multi-scale relationship for drug response prediction. MultiDRP can fuse both internal correlation of feature items and external relationship of biomedical entities by hierarchically integrating graph attention and self-attention networks to improve the anticancer drug response prediction. A variety of results showed that MultiDRP generated the great representation by integrating multi-scale relationships, and achieved higher performance compared to existing methods on various prediction scenarios. The results of network proximity, gene ontology biological process (GOBP) enrichment, and drug pathway association analysis show that MultiDRP can accurately screen the sensitive and resistant drugs for cancer cell lines. *In vitro* experiments, eight novel drugs predicted by MultiDRP exhibited high sensitivity to lung cancer cell line NCI-H23, seven of which showed IC₅₀ values of less than 10 nM. These results further suggest that MultiDRP can serve as a powerful tool for anticancer drug response prediction. The source data and code are available at <https://github.com/pengsl-lab/MultiDRP.git>.

1. Introduction

Cancer is a highly heterogeneous disease and a leading cause of death worldwide. With strong inter-patient heterogeneity, cancers exhibit high levels of genetic and phenotypic variability [1,2]. As a result of heterogeneity, patients with the same type of cancer respond differently to the same drug therapy [1,3]. Therefore, drug response prediction (DRP) in cancer patients is essential and meaningful for precision medicine [3]. In other words, one of the critical goals in precision medicine is prediction of anticancer drugs by using the genetic information of patients. Unfortunately, the lack of drug response data at the patient level limits precision treatments for cancer patients [4,5], which has motivated the creation of large-scale preclinical data resources, such as the Profiling Relative Inhibition Simultaneously in Mixtures (PRISM) dataset [6] and the Genomics of Drug Sensitivity in Cancer (GDSC) dataset [7]. In these studies, a wealth of drug response data in different cell lines and multi-omics data are generated by high-throughput technologies. These data provide vital

information for building computational models for anticancer drug response prediction [8,9]. Therefore, computational technique-based drug response prediction at the cell line level has been the foundation for the development of personalized treatment and has become an important step in drug discovery [3,9,10].

In the past decade, a range of machine learning approaches, such as, support vector machine [11] and logistic regression [12], have been widely applied to the field of drug response prediction and achieved promising prediction performance [13]. In addition, more advanced models, such as multi-task learning [14], matrix factorization [15, 16], and ensemble learning mechanisms [17] have been developed to further improve the performance of drug response prediction. However, most machine learning-based methods tend to rely on biomedical feature engineering. Unfortunately, feature engineering in the biomedical field exhibits a low signal-to-noise ratio [18], thus limiting the performance of machine learning-based drug response prediction. Recently, deep learning as a subclass of machine learning approaches,

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has emerged as a leading paradigm for drug discovery [19,20]. A key advantage behind deep learning-based drug response prediction methods is that deep learning algorithms can capture complex nonlinear input–output relationships and automatically learn high-quality representations from low-level multi-omics data [21–23]. According to the format of the input data, deep learning-based drug response prediction methods can be roughly divided into two major categories: feature descriptor-based approaches and network-based approaches. In feature descriptor-based methods [24], all drugs and cell lines are represented by a continuous value (e.g., gene expression and physicochemical features) or discrete value (e.g., somatic mutation and molecular fingerprints), and then these continuous or discrete features are fed into deep learning models including deep neural networks or convolutional neural networks for drug response prediction. These feature descriptor-based methods achieve better performance compared to machine learning models for drug response prediction. However, most feature descriptor-based methods fail to capture the relationship among biomedical entities. These relationships among entities imply biomedical mechanisms and play an important role in drug development. For example, in a drug similarity network, each edge implies that two end-drugs tend to share similar therapeutic properties [25–27].

To alleviate the above-mentioned limitations, a few studies have explored network-based deep learning for drug response prediction [28, 29]. The core idea of network-based approaches is to construct similarity networks (e.g., structural similarity among drugs and similarity of copy number variation profiles among cell lines) or biomedical heterogeneous networks (e.g., drug–protein–cell line networks) that can naturally simulate the complex interactions in living systems. Subsequently, these network data are fed into deep learning models for drug response prediction. In particular, the graph neural network (GNN), which is a novel deep learning architecture specifically designed for graph structure data [30], has emerged as the most popular technology in network-based drug response prediction. Concurrently, GNNs learn biomedical entity representations by aggregating information from their neighborhoods in associated networks, thus modeling the relationship among entities. In other words, network-based methods can capture the external relationship among biomedical entities compared to feature descriptor-based approaches. In addition to the external relationship among biomedical entities, in fact, most feature items in entities (e.g., genes in cell lines) interact with each other. For example, there are complex interactions among genes in cell lines [31,32] and associations among substructures in molecules. This kind of internal correlation of feature items in cell lines or drugs provides microcosmic information for drug response prediction. Therefore, incorporating the internal correlation of feature items in entities may be helpful to improve drug response prediction and provide new insights into drug discovery. However, most network-based methods ignore the internal correlation of feature items in cell lines or drugs. Therefore, it is challenging to develop a deep learning approach that can capture both the external relationship among biomedical entities and the internal correlation of feature items to improve drug response prediction.

Here, we propose MultiDRP that develops the hierarchical attention networks integrating multi-scale relationship to improve anticancer drug response prediction. In this framework, the cell line and drug networks are constructed based on copy number variation profiles and physicochemical properties, respectively, to simulate the association among biological entities. Therefore, MultiDRP proposes a biomedical network-driven graph attention model to capture the external relationship among cell lines and the relationship among drugs. Network-based attention simulates the different contributions of various neighbors to central biomedical entities. Moreover, a multi-head self-attention network is used to learn the association among feature items in entities, such as the relationship among genes in cell lines and the relationship among substructures in molecular fingerprints. The self-attention network models the long-range dependencies among different

feature items. Overall, MultiDRP simultaneously captures the external relationship among entities and the internal association among feature items to improve drug response prediction. For various prediction scenarios based on known drug response data, a mass of results suggest that MultiDRP achieved better performance and stronger robustness than six baseline methods. The high-quality representations generated by MultiDRP could greatly improve the performance of machine learning-based drug response prediction. Furthermore, based on network proximity, GOBP enrichment, and drug pathway association analysis, we found that there was a significant difference between our predicted sensitivity and resistance drugs, further demonstrating that MultiDRP has great discrimination ability. Finally, for eight novel drugs that are sensitive to the lung cell line NCI-H23, we performed *in vitro* experiments to validate the practical applicability of MultiDRP. We observed that seven of eight drugs exhibited an IC_{50} value of less than 10 nM. In summary, MultiDRP is a powerful tool to predict drug response.

2. Materials and methods

2.1. The overall framework of MultiDRP

MultiDRP is an end-to-end deep learning framework for anticancer drug response prediction, which can fuse the internal correlation of feature items in biomedical entities and external relationship among entities by hierarchically integrating graph attention (GAT) network [33, 34] and self-attention models [35,36], respectively, as shown in Fig. 1(A). There are three key components: network construction, representation learning, and drug response prediction. For network construction, we first use principal component analysis (PCA) [37] to reduce the dimensionality of cell line characteristics (i.e., gene expression profiles and copy number variation profiles). Then, the vectors of copy number variation profiles are used to construct association networks among cell lines, and vectors of gene expression profiles are treated as the initial features of cell lines. Analogously, the physicochemical properties and molecular fingerprints are also utilized to construct drug association networks and initial features, respectively. In representation learning, the encoder is composed of a stack of identical layers. Each layer has two sub-layers. The first is a graph attention (GAT) layer that aims to capture the relationship among cell lines or drugs. The second is a multi-head self-attention layer that focuses on exploring internal correlation of items in gene expression profiles and molecule fingerprints. We employ a residual connection [38] around each of graph attention layer and self-attention layer to fusion the external and internal representations, followed by layer normalization (LN) [39] the representations of cell lines and drugs are concatenated and then fed into the multilayer perceptron (MLP) to predict drug response. To validate the performance of MultiDRP, we first executed an evaluation based on all known drug response data. Next, we compared the mechanisms of sensitive and resistant drugs predicted by MultiDRP. Finally, the predicted sensitivity drugs were executed for the *in vitro* validation on lung cancer cell line NCI-H23.

2.2. Dataset and data preprocessing

2.2.1. Dataset construction

Drug response data: In this study, drug response value were from PRISM dataset [5] that was developed by Broad Institute of MIT and Harvard to evaluate the anticancer effects of non-oncology drugs on human cancer cell-lines [6,40]. By testing the growth inhibitory activity of drugs in human cancer cell lines, PRISM obtained high-quality large-scale drug response data which include the omics data of cell lines, features of drugs, and half-maximal inhibitory concentration (IC_{50}) value which is a measure of the sensitivity of cell lines to drugs. Generally, a lower IC_{50} value indicates a higher sensitivity of a cell line to a given drug. Importantly, PRISM encompasses a large number

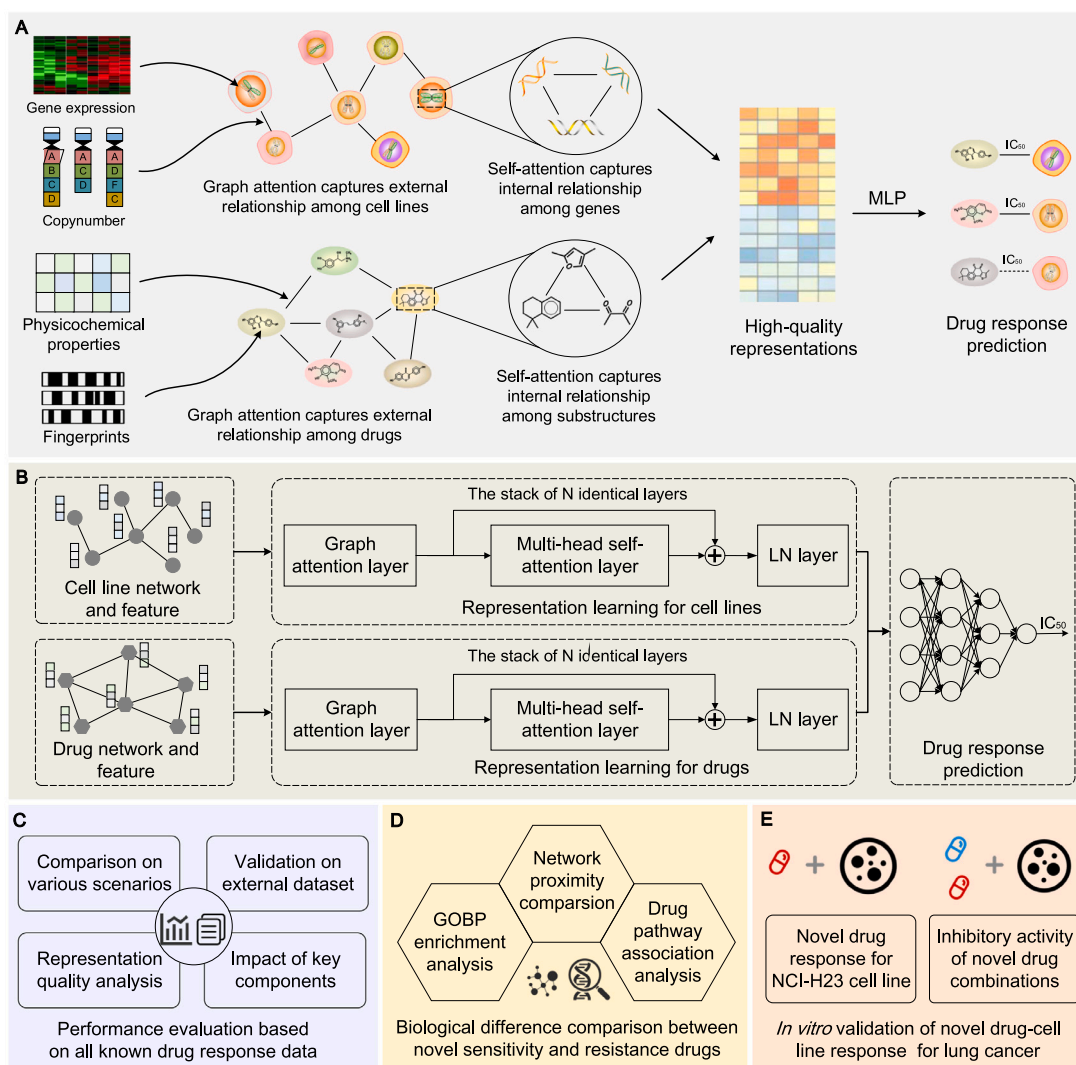


Fig. 1. The overview of this study. (A) Schematic workflow of MultiDRP. (B) Hierarchical attention network architecture. The association networks and initial features are fed into the hierarchical attention network which can capture the external relationship among entities and internal correlation of feature items. (C) Performance evaluation based on all known drug response data. (D) Biology difference comparison between predicted sensitivity and resistance drugs. (E) *In vitro* validation of novel drugs for lung cell line NCI-H23.

of drugs across a diverse panel of genomically characterized cell lines, allowing to capture the molecular diversity inherent in human cancer. In this work, we filter out abnormal samples (1) whose IC_{50} values exceed the lower or upper extremes of a box-plot, and (2) whose omics features of cell lines is incomplete. In addition, we calculate the logarithmic IC_{50} value (i.e., $\log(IC_{50})$) as labels. Finally, we obtained 274,279 log-transformed IC_{50} values between 1391 drugs and 455 cell lines.

Drug data: We selected molecular fingerprints and physicochemical properties to describe drugs because these features can provide rich information about chemical structure and characteristics. The molecular fingerprints and physicochemical properties of drugs were widely used in deep learning-based drug discovery. For each drug, the chemical structure in SDF format was extracted from the PubChem [41], and then converted into molecular fingerprints through ChemmineR package [42]. The molecular fingerprints are 881-dimension bit vectors that represent the presence or absence of particular substructures in each drug. Nevertheless, physicochemical features are a series of continuous values that represent physical or chemical properties quantitated via ChemmineR, Pybel [43], RDKit [44], and JOELib2 (<http://www.ra.cs.uni-tuebingen.de/software/joelib/index.html>).

Cell line data: We used gene expression data and the DNA copy number data to describe cell lines. First, gene expression data reflects

the activity level of genes within a cell, providing insights into the underlying biological processes and pathways that may influence drug response. Second, gene expression data captures the expression level of thousands of genes, and offers a comprehensive view of the molecular characteristics of cell lines. Third, previous research has demonstrated that gene expression data plays a key role in drug response prediction [45]. The copy number data offers the characteristics at gene level for cell line. Previous research has highlighted the prevalence of copy number alterations in cancer, many of which confer disadvantageous effects [46]. These alterations play pivotal roles in cancer initiation, progression, and predisposition [47]. In this work, the gene expression and copy number data were collected from CCLE database [48].

2.2.2. Data preprocessing

In real drug discovery, the characteristic data of drugs and cell lines are difficult to directly apply to deep learning because of their high dimensionality. For example, the dimensional size of gene expression profiles in each cell line is 48,392. Therefore, all characteristic data of drugs and cell lines are first mapped into low-dimensional space by PCA, which is an unsupervised machine learning technique. Then, we computed the cosine similarity among all cell lines by low-dimensional copy number variation data. Each cell line and drug has a different number of neighbors in real biological networks. However,

it is challenging to select the exact number of neighbors for each biological entities. Inspired by previous studies including ADRML [49], MinDrug [50], DSPLMF [51], Auto-HMM-LMF [52], TCLMF [53], and CDSML [54], for each cell line, the 10 most similar objects are chosen as its neighbors, that is, all cell lines construct a binary network. In addition, the low-dimensional gene expression data are treated as the initial features of cell lines. Similarly, we use the fingerprint and physicochemical properties to generate the association networks and initial features of drugs, respectively. Finally, the association networks and initial features of cell lines and drugs are fed into the hierarchical attention network for drug response prediction.

2.3. Hierarchical attention networks integrating multi-scale relationships

2.3.1. Graph attention network capturing the external relationships

The cell line networks and drug networks are fed into different GAT [33] modules for capturing the external relationships among biomedical entities, respectively. GAT, a novel graph neural network, aggregates neighbor features via attention coefficients to learn representations of nodes in network data. The attention coefficients denote the contribution of neighbors to a central node, and are learned by masked attention mechanisms that was specifically proposed for graph structured data. In mask attention models, we only compute the relationship between center nodes and its neighborhoods in graph. Intuitively, the mask attention models mainly focus on the topological structures in graph. Formally, the attention coefficient α_{ij} between nodes v_i and v_j is obtained as follows:

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(\bar{a}^T \left[\mathbf{W}\bar{h}_i \parallel \mathbf{W}\bar{h}_j\right]\right)\right)}{\sum_{j \in N_i} \exp\left(\text{LeakyReLU}\left(\bar{a}^T \left[\mathbf{W}\bar{h}_i \parallel \mathbf{W}\bar{h}_j\right]\right)\right)} \quad (1)$$

where N_i is all neighborhoods of node v_i in network, $\bar{a}$ is a trainable parameter matrix, $\mathbf{W}$ is a trainable weight matrix mapping the input features to the hidden space, and $\parallel$ represents concatenation operations.

Similar to Transformer [35], GAT employs multiple masked attention mechanisms to improve the model stability and representation capability. Specifically, GAT utilizes K independent attention models to compute the features of each node, and then these features are concatenated and again projected. Therefore, the final output features of nodes can be calculated by:

$$\bar{h}_i' = \mathbf{W}_{pro} \parallel_{k=1}^K \sigma\left(\sum_{j \in N_i} \alpha_{ij}^k \mathbf{W}^k \bar{h}_j\right) \quad (2)$$

where α_{ij}^k is an attention coefficient of the k th masked attention mechanism, $\mathbf{W}^k$ is the corresponding weight matrix of the input linear transformation, $\sigma(\cdot)$ is a nonlinear function, and $\mathbf{W}_{pro}$ is a projection matrix that aims to map the representations from multiple subspaces into the same space.

2.3.2. Self-attention network reinforcing the internal correlation

We use GAT layers to capture the external relationships among entities. However, GAT layers ignore the internal relationship in feature items in biomedical entities, such as, the relationship among genes in cell lines. In addition, GAT is easy to lead the over-smooth situation that refers to all nodes with the high-similarity representation. To relieve the issues, this work integrates a self-attention layer after each GAT layer.

The self-attention model [35] can be described as mapping a query $\mathbf{Q}$ and a set of key-value pairs $(\mathbf{K}, \mathbf{V})$ to an output vector, where $\mathbf{Q}, \mathbf{K}$ and $\mathbf{V}$ are attained by linear transformation of $\mathbf{H} = \{\bar{h}_1', \dots, \bar{h}_l'\}^T$ that is from the GAT layer. More specifically, the output vectors are a weighted sum of the values, as follows:

$$\mathbf{H}' = \left(\text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V} \right)^T \quad (3)$$

where d_k is the size of $\mathbf{K}$. The masked attention mechanism is the extension of self-attention model in Transformer [35]. The original self-attention can learn the dependencies among arbitrary elements, such as each token in a sentence. Therefore, the original self-attention is equal to model a fully-connected graph where each element is regarded as a node.

Similar to the GAT layers, we adopt multiple self-attention mechanisms. Therefore, the output of multiple self-attention with different queries, keys and values is added. Multi-head attention guides the model to jointly attend to information from different subspaces at different position items.

2.4. Residual connections and layer normalization

In MultiDRP, we employ residual connections [38] around each GAT layer and self-attention layer. In this work, residual connection, which is equal to skip connection, aim to directly adding the representation from graph attention layer into representation from self-attention layer, thus fusing the representations of internal and external space. Simultaneously, residual connection is able to relieve the gradient vanishing phenomenon in training process. In addition, we use layer normalization [39] after residual connections to improve the speed of model convergence and robustness. For single sample, layer normalization is aimed to normalize all of the neuron values in a hidden layer (as described in Supplementary Text S1). Layer normalization can reduce the difference between various training samples, thus improving training speed and model performance.

2.5. Drug response prediction and optimization

To predict drug-cell line responses, the representations of cell lines and drugs are concatenated and then fed into the multilayer perceptron with three fully connected layers. There is no activation function in the output layer. The hidden layers adopt the ReLU activation function. In this work, drug response prediction is treated as a regression question that aims to fit the IC_{50} values of the cell line-drug response. Therefore, we use the mean square error as the loss function, and Adam as the optimizer for updating model parameters.

2.6. Measure algorithms of differences between sensitive and resistant drugs

For the sensitive and resistant drugs predicted by MultiDRP, we calculated network proximity [55] to compare their topological associations with diseases in the protein-protein interaction (PPI) network. Simultaneously, we calculated the association score between each drugs and pathways [50,51,56] to further analyze the differences between sensitive and resistant drugs.

2.6.1. Network proximity measure

Network-based proximity is performed to measure the topological association between drugs and lung cancer in protein-protein interaction network and identify drug combinations. The network-based drug-disease proximity is defined as follows:

$$d(X, Y) = \frac{1}{|X| + |Y|} \left(\sum_{x \in X} \min_{y \in Y} d(x, y) + \sum_{y \in Y} \min_{x \in X} d(x, y) \right) \quad (4)$$

where X represents the set of targets related to a drug, and Y represents the set of proteins related to a disease. $|X|$ and $|Y|$ are the size of the drug target set and the disease protein set, respectively, and $d(x, y)$ is the shortest distance between proteins x and y in PPI network. Here, we use the proteins related to lung cancer identified by GPSnet [55], a novel network-based disease module identification methodology, as the lung cancer-specific protein set in PPI network.

To assess the significance of the distance between a drug and a disease, the network proximity is converted to Z-score by permutation tests:

$$Z_{d(x,y)} = \frac{d(X,Y) - \mu_t}{\sigma_t} \quad (5)$$

where μ_t and σ_t are the mean and standard deviation of a reference distance distribution, respectively, which is obtained by a permutation test that is repeated 1000 times. In each permutation test, two randomly selected protein lists with similar degree distributions to those of X and Y are used to calculate the shortest distance in PPI network. Generally, closer proximity in PPI network represents a stronger association between a drug and a disease. This is mainly because a drug with high sensitivity to a disease, its targets should be within or in the immediate neighborhoods of the disease-related proteins in PPI network [55,57,58].

To identify effective drug combinations, we need to not only quantify the topological relationship between drugs and disease, but also detect the overlap degree of drug pairs. Therefore, network-based separation between drug A and drug B is defined as:

$$S(A,B) = \langle d(A,B) \rangle - \frac{\langle d(A,A) \rangle + \langle d(B,B) \rangle}{2} \quad (6)$$

where $\langle d(\cdot, \cdot) \rangle$ are the average shortest path lengths within the drug-target module. For $S(A,B) > 0$, the targets of two drugs are topologically separated. If the target modules of two drugs are separated in PPI, but close to the proteins related to a disease, the combinations of these two drugs are effective for treatment of this disease.

2.6.2. Association between drugs and pathways

To obtain the association between drugs and pathways, we first calculate the pathway activity score based on the gene expression of cell lines. According to the guidance [50], we selected Biocarta signaling pathways from MSigDB [59]. If more than 10% of genes in a pathway have no gene expression data in CCLE, we would exclude the pathway. The activity score between the cell line C_i and the selected pathway P_j is calculated as follows:

$$S(C_i, P_j) = \sum_{g_k \in P_j} (\log_2(e(C_i, g_k)) - \text{MED}(\log_2(e(C, g_k)))) \quad (7)$$

where $e(C_i, g_k)$ represents the gene expression of gene g_k in cell line C_i , $e(C, g_k)$ represents the gene expression of gene g_k in all cell lines C , and $\text{MED}(\cdot)$ represents the calculation of the median value.

Based on the activity score between pathways and cell lines, we then define the association score between drug D_i and pathway P_j as:

$$AS(D_i, P_j) = \text{Pearson}(\overline{D_i}, \overline{P_j}) \quad (8)$$

where $\text{Pearson}(\cdot)$ denotes the Pearson correlation coefficient (PCC), $\overline{D_i}$ represents the predicted IC_{50} value of drug D_i in various cell lines, and $\overline{P_j}$ represents the activity scores of pathway P_j in various cell lines, that is, $\overline{P_j} = \{S(C_1, P_j), S(C_2, P_j), \dots, S(C_n, P_j)\}$ where n is the number of cell line. For a specific tissue such as lung cancer, the association between a drug and a pathway can be measured by PCC of the predicted IC_{50} values of drug in all lung cancer cell lines and the activity scores of the pathway in all lung cancer cell lines. The greater the absolute value of the association score, the stronger correlation between the drug and pathway in lung cancer cell lines.

2.7. Cell viability assay

Cells and cell culture: In order to verify the practicability of MultiDRP for specific cell lines, we performed *in vitro* experiment assays based on NCI-H23 cell lines that were obtained from Procell Life Science & Technology, Wuhan, China. Cell lines were validated by STR profiling that is performed by Cell Line Authentication Service at ATCC. Cells were cultured in RPMI1640 (#SH30809, Hyclone) +10% FBS

(#10500064, Gibco) + penicillin-streptomycin (#15070063, Gibco) medium.

Cell viability and drug combination assays: For the cell viability assay, NCI-H23 cells (5000 cells/well) were seeded in 96-well plates (100 μL per well) at to adhere overnight, and then treated with different drug doses for 72 h, and dimethyl sulfoxide (DMSO) was used as control. Subsequently, according to the manufacturer's instructions (Dojindo Laboratories, mamoto Ken, Japan), cell viability was evaluated by the cell counting kit-8 (CCK-8 Kit). We used Graphpad 7.0 to calculate the IC_{50} value. For drug combination assays, the Chou-Talalay equation was utilized to quantitatively define the synergistic effect of drug pairs [60]. Combinational Index (CI) <1 indicates synergism; CI > 1 indicates antagonism. NCI-H23 cells (5000 cells/well) were seeded in 96-well plates (100 μL per well) to adhere overnight, and then cells were treated with different concentrations of two single drugs and combination treatment for 72 h, respectively. Subsequently, cell viability was evaluated with a CCK-8 kit according to the manufacturer's instructions (Dojindo Laboratories, Japan).

3. Results

3.1. Great performance on different prediction scenarios

To demonstrate the performance of our methods, we compared MultiDRP with **six baseline approaches** in this experiment, which includes KNN [61], DecisionTree [62], GBDT [63], DeepDSC [64], DeepDRK [28], and GraOmicDRP [65]. A detailed description of baselines can be found in Supplementary Text S2. In this work, the root mean squared error (RMSE), mean absolute error (MAE), Pearson's correlation coefficient (PCC), and R-squared (R^2) are treated as metrics to evaluate MultiDRP and all baselines. We first performed the cross-validation experiment, in which all known cell line-drug response samples were randomly split into a training (80%), validation (10%), and testing sets (10%). The training and validation sets were used to train models and tune parameters. **MultiDRP and baselines were repeatedly run 10 times, and the average results were calculated.** As shown in Fig. 2(A) and Supplementary Figure S1, we observed that MultiDRP outperformed five baselines on all metrics. Interestingly, we found that MultiDRP generated the limited improvements to DeepDRK in terms of PCC. However, MultiDRP yielded an approximately 5% improvement in terms of RMSE, MAE, and R^2 compared to DeepDRK. These results highlight the great capability of MultiDRP in the precision prediction of anticancer drug response. In addition, another interesting finding was that the average results of deep learning methods (i.e., MultiDRP, DeepDSC, DeepDRK, and GraOmicDRP) are often superior to those of machine learning approaches for drug response prediction. In particular, KNN (RMSE = 1.277, MAE = 0.999, PCC = 0.233, R^2 = 0.010) seemed to completely lose the predictive ability. A possible explanation for this result is that machine learning models heavily depend on biomedical feature engineering. Unfortunately, the biomedical features with a low signal-to-noise ratio limit the performance of machine learning models.

In the above experiments, the training samples were much larger than the tested samples. However, there are only a handful of known drug response values in real-world drug development scenarios. To simulate the practical drug prediction scenario, we randomly selected a handful of known drug-cell line samples (i.e., 5%, 10%, 15%, and 20%) to train MultiDRP and five baselines. The rest of the samples are treated as the test set to evaluate the performance of various models. As shown in Fig. 2(B) and Supplementary Figure S2, with the growth of training data size, the predictive performance of all methods was improved gradually on all evaluation metrics. Interestingly, GBDT was superior to deep learning models (i.e., GraOmicDRP, DeepDRK and DeepDSC) and became the second best method when there were only 5% training samples. Concurrently, GBDT and DecisionTree generated marginal improvements when the volume of the training set increased

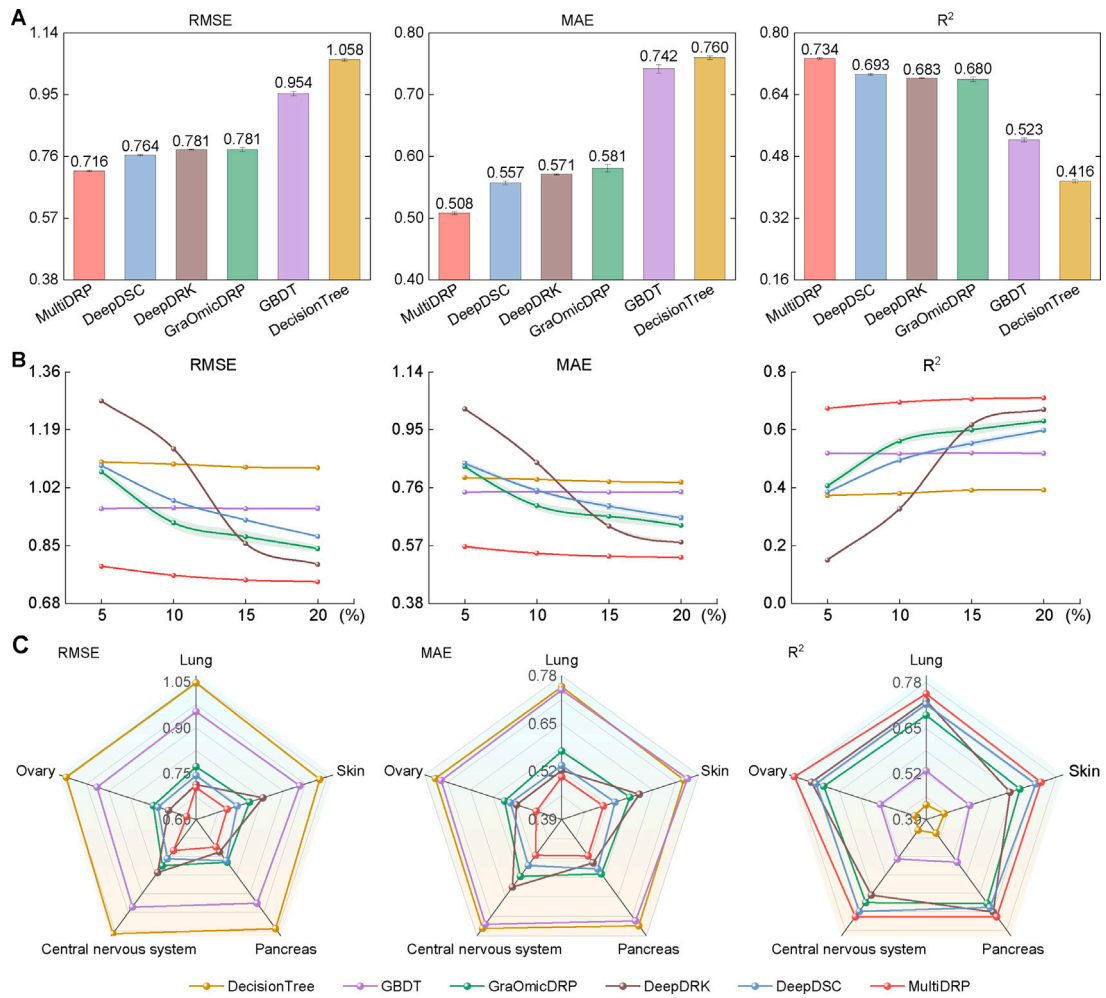


Fig. 2. Results of MultiDRP and baselines on different drug prediction scenarios where the error bars and shaded areas denote the standard deviation values that are calculated across ten results. (A) Results on large-scale labeled samples. (B) Results on few-shot prediction. (C) Results on cell lines of specific tissues.

from 5% to 20%. This result may be attributed to the fact that GBDT and DecisionTree have the smaller parameter sizes and are easier to saturate when compared to deep learning methods. More importantly, we observed that the performance of all methods reduced when compared to the results in Fig. 2(A). In other words, all of methods did not perform as well because the volume of the training set was reduced. In particular, DeepDRK was reduced by 38.3%~52.0%, when the ratio of training sets decreased from 20% to 5%. However, the performance of MultiDRP did not exhibit much fluctuation and was superior to five baselines when there was a small number of training samples. These results suggest that most existing methods are prone to be influenced when be applied to sparse data, while MultiDRP can partly overcome this limitation and relieve the requirement of label size.

In this study, there are the different numbers of cell lines in various tissues, as shown in Supplementary Figure S3. For a given drug, there is a large sensitivity difference among cell lines in diverse types of tissues. To evaluate the performance of all methods on different types of tissues, we tested them on five types of tissues with the most cell lines, i.e., lung, skin, pancreas, central nervous system, and ovary. Specifically, 10% of drug-cell line samples in a given tissue were treated as testing sets. The rest of the samples were used as training and validation sets. In Fig. 2(C) and Supplementary Figure S4, we found that MultiDRP consistently achieved the best performance on five tissues. Another phenomenon was that DeepDRK tended to achieve a large margin difference among various types of tissues compared to other methods. In contrast, MultiDRP achieved the excellent and consistent results on different tissues.

3.2. Superior performance on GDSC and TCGA datasets

In this experiment, we evaluated MultiDRP and baselines on two datasets, i.e., GDSC [7] and The Cancer Genome Atlas (TCGA) [66] as described in Supplementary Text S3. Note that drug response prediction in TCGA dataset is a binary classification task. Therefore, we adopted the area under receiver operating characteristic (AUROC) curve and the area under precision recall (AUPR) as evaluation metrics on TCGA dataset. On GDSC dataset, we found that MultiDRP achieved the best results on all evaluation metrics (RMSE = 1.038, MAE = 0.781, PCC = 0.899, R^2 = 0.806) as shown in Fig. 3(A). Similarly, MultiDRP (AUROC = 0.913, AUPR = 0.937) was superior to the other five methods, with 9.4% average improvement in AUPR and 8.9% average improvement in AUROC. Interestingly, for GDSC and TCGA, DeepDRK exceeded DeepDSC and became the second best method. However, MultiDRP demonstrated strong generalization ability and achieved the best performance on various datasets.

3.3. High-quality representations of biomedical entities

In the above experiments, KNN, GBDT, and DecisionTree showed lower performance compared to deep learning models. We conjectured that the worse performance is because of the biomedical features with a low signal-to-noise ratio. Therefore, we further studied the effect of input features on machine learning models. Specifically, after the training stage, MultiDRP generates the low-dimensional representation

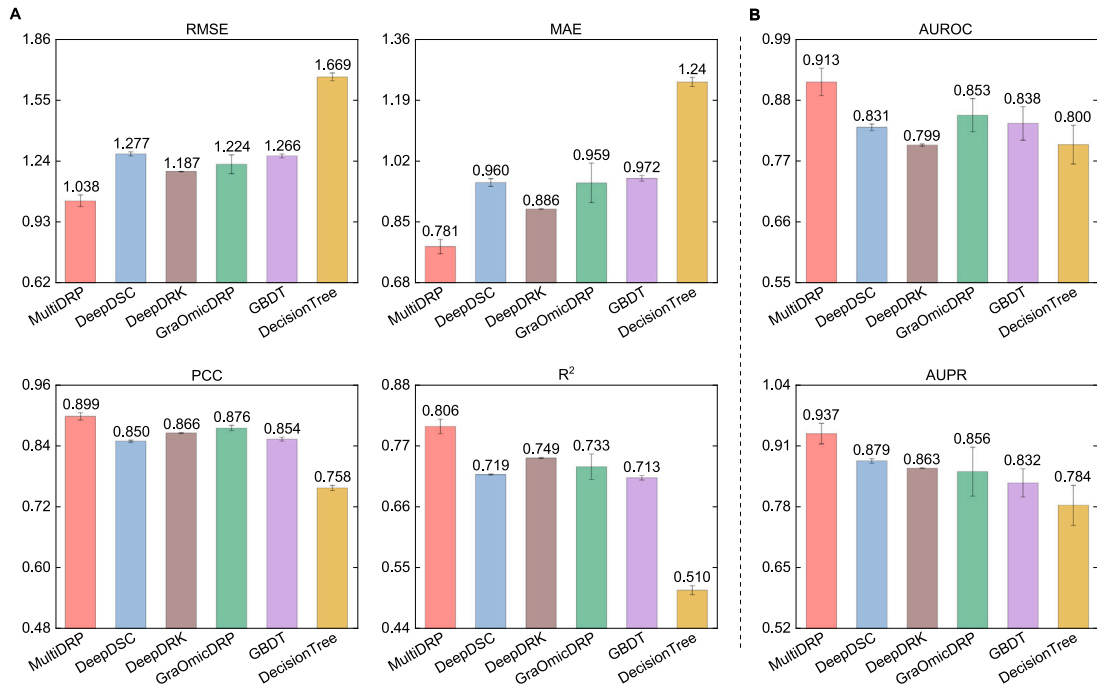


Fig. 3. Results on GDSC and TCGA dataset where the error bars denote the standard deviation values that are calculated across ten results.

Table 1
Results of machine learning with different representations.

Method	RMSE $\pm$ std	MAE $\pm$ std	PCC $\pm$ std	R ² $\pm$ std
KNN	1.277 $\pm$ 0.038	0.999 $\pm$ 0.030	0.233 $\pm$ 0.069	-0.105 $\pm$ 0.007
Multi-KNN	0.773 $\pm$ 0.001	0.557 $\pm$ 0.002	0.832 $\pm$ 0.001	0.689 $\pm$ 0.001
DecisionTree	1.058 $\pm$ 0.004	0.760 $\pm$ 0.003	0.712 $\pm$ 0.002	0.416 $\pm$ 0.004
Multi-DecisionTree	1.029 $\pm$ 0.003	0.735 $\pm$ 0.002	0.729 $\pm$ 0.001	0.450 $\pm$ 0.003
GBDT	0.954 $\pm$ 0.007	0.742 $\pm$ 0.007	0.761 $\pm$ 0.003	0.523 $\pm$ 0.005
Multi-GBDT	0.723 $\pm$ 0.001	0.516 $\pm$ 0.002	0.854 $\pm$ 0.001	0.729 $\pm$ 0.003

vectors of drugs and cell lines and then these feature vectors were fed into KNN, DecisionTree, and GBDT for drug response prediction, named Multi-KNN, Multi-DecisionTree, and Multi-GBDT, respectively. Finally, the results of test set are shown in Table 1 where the best results are marked in **boldface**. We observed that Multi-KNN, Multi-DecisionTree, and Multi-GBDT were superior to KNN, DecisionTree, and GBDT on all evaluation metrics. The results suggest that MultiDRP generated the high-quality representations that can improve performance of machine learning models.

To further verify the effectiveness of the encoder of MultiDRP, we utilized *t*-SNE algorithm [67] to visualize the sample distribution based on the initial feature preprocessed via PCA and representation generated via MultiDRP, respectively. To be specific, we randomly selected 1% of drug-cell line pairs in testing set and observed the sample distribution according to embedding of different stages. The initial features of cell lines and drugs were concatenated and mapped into a two-dimensional space by *t*-SNE algorithm. For the visualization of sample distribution, the value of IC₅₀ was used to color the drug-cell line pairs as shown in Fig. 4(A). Similarly, the representations generated by MultiDRP are mapped to a two-dimensional space and colored as shown in Fig. 4(B). We observed that drug-cell line pairs with different IC₅₀ values were completely mixed in two-dimensional space of initial feature. In contrast, in representation space of MultiDRP, drug-cell line pairs were separated according to their IC₅₀ values. As shown in Figs. 4(C) and 4(D), we observed the same results on five kinds of tissues with the most cell lines, i.e., lung, skin, pancreas, central nervous system, and ovary. In this experiment, 10% of drug-cell line samples in a given tissue were treated as testing sets. The rest of the

samples were used for training and validation sets. Next, we selected 1% drug-cell line pairs in testing set and observed their distribution. These results indicate that MultiDRP is able to efficiently generate high-quality representation vectors for drug response prediction and maintain a consistent performance on different tissues.

3.4. Superior performance for cell line blind test

3.4.1. Performance comparison for cell line blind test

In drug discovery of real world, it is necessary to predict the sensitive drugs for new cell lines. Therefore, we designed a blind test experiments in which the cell lines in test sets are unseen in training sets [56,68]. The blind test can simulate the drug response prediction for new cell lines and more challenging. To be specific, we randomly select 10% cell lines and then all drug response samples associated with these cell lines are treated as test set. The rest of cell lines and samples are used for training models. To ensure a fair and systematic evaluation, we compared MultiDRP with five state-of-the-art methods, including ADRML [49], MinDrug [50], DSPLMF [51], AutoHMM-LMF [52], CaDRReS [56]. These methods are widely tested on various datasets and have achieved great performance. The details can be found in Supplementary Text S4. Results of MultiDRP and baselines are shown in Fig. 5. We observed that MultiDRP outperformed five baselines on multiple metrics, achieving a 36.9%, 31.04%, and 14.32% average improvement on RMSE, MAE and PCC value. Interestingly, MinDrug achieves the competitive value in terms of RMSE and MAE. However, its PCC value is lower than CaDRReS. Simultaneously, we also noticed that CaDRReS generates the slightly higher PCC value

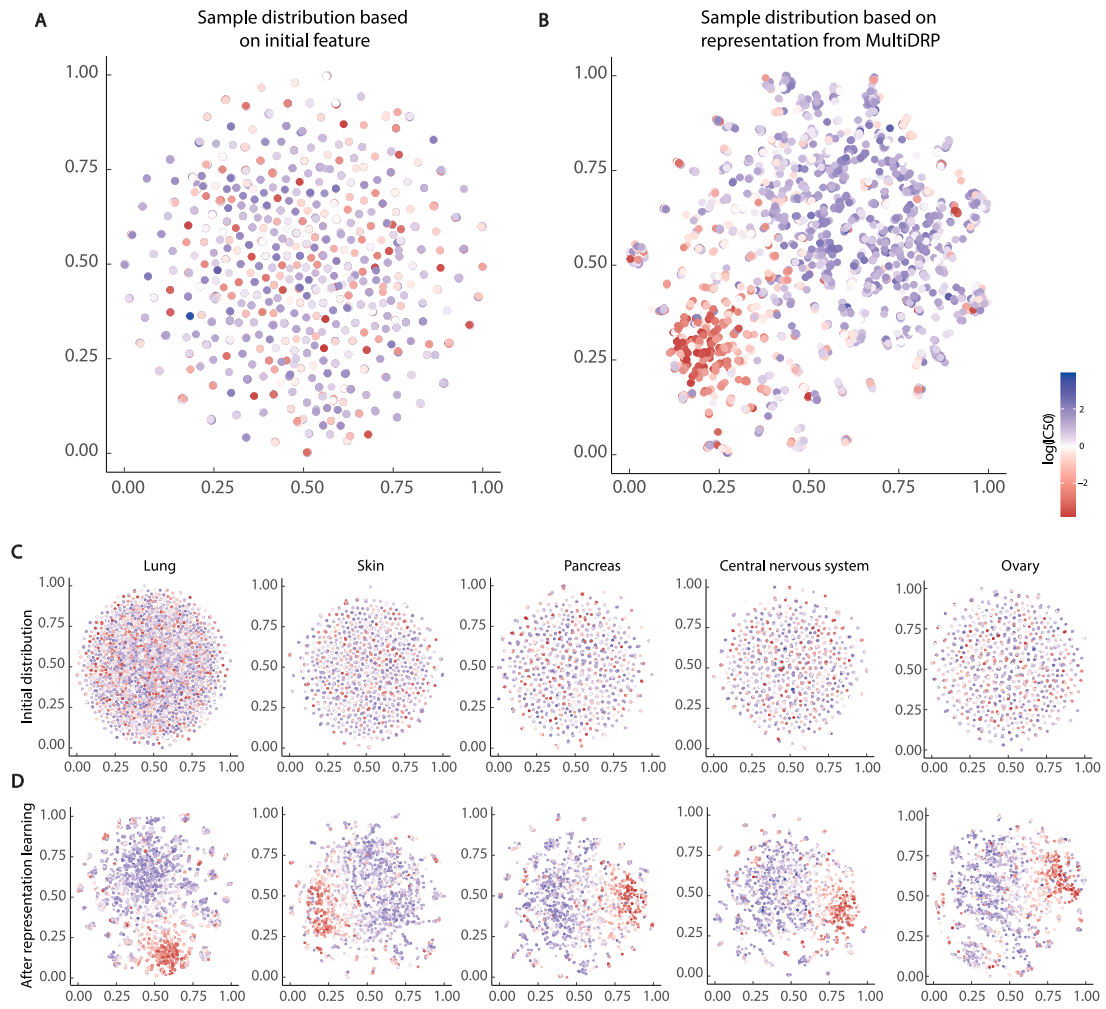


Fig. 4. The distribution of drug-cell line pairs based on different representations of cell lines and drugs. (A) Sample distribution based on initial features preprocessed by PCA. (B) Sample distribution based on representation vectors generated by MultiDRP. (C) Initial feature-based sample distribution on five specific tissue types. (D) Sample distribution on five specific tissue after MultiDRP representation learning.

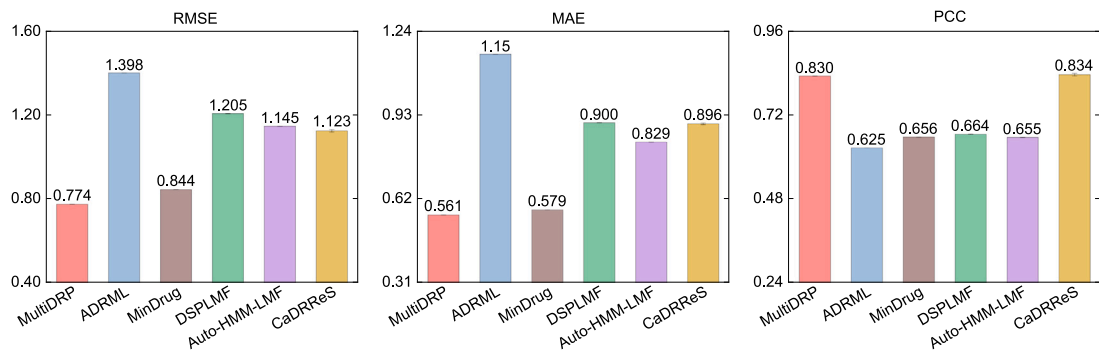


Fig. 5. Results of MultiDRP and baselines for cell line blind test.

than our method. However, MultiDRP yielded an approximately 35% improvement in terms of RMSE and MAE when comparing to CaDRReS. In addition, CaDRReS generated relatively higher standard deviation value compared to other methods. These results suggest that MultiDRP can achieve higher and robust performance for cell line blind test.

3.4.2. Representation quality comparison for cell line blind test

To verify the representation of different models, we compared the representation of MultiDRP and five baselines for cell line blind test.

It is worth noting that we only select GBDT to verify the representation ability of MultiDRP and five baselines, named Multi-GBDT, ADRML-GBDT, MinDrug-GBDT, AH-LMF-GBDT, DSPLMF-GBDT, and CaDRReS-GBDT, respectively. Because GBDT achieved the best performance compared other machine learning models in cross-validation experiments. To be specific, after the training stage, MultiDRP and five baselines generated the representation vectors of drugs and cell lines, and then these features in training sets were used to train GBDT model. Finally, samples in test set was used for evaluating the representation performance. For cell line blind test, the results of GBDT with

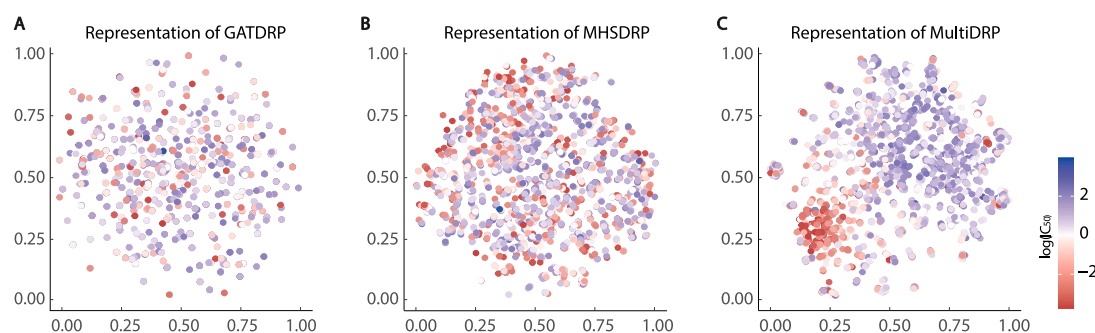


Fig. 6. The distribution of drug-cell line pairs based on representations of attention networks at different hierarchies.

different representations is shown in Supplementary Materials Table S1. We observed that Multi-GBDT achieved the best performance on all evaluation criteria when compared five baselines. In particular, in terms of RMSE values, MultiDRP-GBDT yielded an approximately 13.19% average improvement compared to baselines. These results further suggested that MultiDRP can generate the high-quality representations, and improve performance of machine learning models for drug response prediction. We also found that MinDrug achieves great results. This phenomenon indicates that it is important to identify the subset of molecules for drug discovery. Theoretically, MinDrug may provide novel insight and guidance for drug response prediction.

Based on cell line blind test, we further compared the representations generated by MultiDRP and five baselines in a two-dimensional space and use IC_{50} value to color samples. As shown in Supplementary Material Figure S5, we observed that drug-cell line pairs with different IC_{50} values were mixed in representation space based on ADRML, MinDrug, and CaDRReS. In contrast, in representation space of MultiDRP, drug-cell line pairs were separated according to the different IC_{50} values. We also noted that drug-cell line pairs with different IC_{50} values were separated, when utilizing the representations generated by Auto-HMM-LMF and DSPLMF.

To quantitatively compare the representations generated by different methods, the median of IC_{50} values was chosen as threshold value according to the guidelines in Auto-HMM-LMF and DSPLMF, and then the drug-cell line pairs in test sets are divided into two classification. Finally, we calculated the Calinski–Harabasz (CH) score [69] and Silhouette Score [70] to evaluate the separation of the two groups. The CH and Silhouette score is generally utilized to assess clustering effectiveness, that is, the higher CH and Silhouette values denote the greater inter-cluster distance and smaller intra-cluster distance. As shown in Table 2, MultiDRP can better differentiate samples from the two clusters, as evidenced by higher CH score and Silhouette score. Interestingly, the representations from MultiDRP, Auto-HMM-LMF, and DSPLMF showed clear sample distribution. However, MultiDRP achieved higher CH score and Silhouette Score. These results further indicate that MultiDRP is able to generate high-quality representation vectors for drug response prediction.

3.5. Ablation analysis

3.5.1. Impact of the internal and external attention networks

MultiDRP integrates graph attention network-based external representation learning and self-attention network-based internal enhancement to capture multi-scale relationships in biomedical entities. To evaluate the contribution of different representation models, we used a GAT network (GATDRP) and a multi-head self-attention network (MHSDRP) to predict drug response, respectively. Similar to MultiDRP, GATDRP and MHSDRP are also end-to-end deep learning models and are trained from scratch for drug response prediction. As shown in Supplementary Materials Table S2, MHSDRP was superior to GATDRP on all evaluation metrics for our dataset. Nevertheless, we found that

Table 2

Comparison of the representations generated by different methods.

Method	CH Score	Silhouette Score
ADRML	5.846	0.003
MinDrug	8.010	0.004
DSPLMF	300.713	0.099
Auto-HMM-LMF	304.636	0.105
CaDRReS	0.133	−0.0003
MultiDRP	375.126	0.119

the performance of GATDRP is slightly better than that of MHSDRP on GDSC dataset. More importantly, we observed that MultiDRP achieved the better performance compared to GATDRP and MHSDRP on the different assessment indicators.

To further illustrate the impact of internal and external relationships in biomedical entities on performance, the representations generated by GATDRP and MHSDRP are mapped to a two-dimensional space and colored as shown in Fig. 6. We observed that drug-cell line pairs with different IC_{50} values were mixed when using the representations generated by GATDRP and MHSDRP. In contrast, when utilizing the representations generated by MultiDRP, drug-cell line pairs were separated according to their IC_{50} values. These results suggest that MultiDRP integrating GAT and multi-head self-attention model was beneficial to improve the drug response prediction. In other words, MultiDRP can improve anticancer drug response prediction by fusing internal and external relationships in biomedical entities.

3.5.2. Selection of similarity functions

To verify the effect of different similarity functions on the performance of MultiDRP, in process of constructing cell line and drug networks, we replaced cosine similarity by Euclidean Distance (MultiDRP-ED), which is also a famous similarity calculation methods. Figure S6 in Supplementary Material summarizes the results of MultiDRP-ED for drug response prediction. We found that there is no significant difference between the performance of MultiDRP-ED and MultiDRP. In particular, the RMSE of MultiDRP-ED and MultiDRP is almost same value. A similar issue can also be observed in a previous study [68]. These results indicate that the different similarity measurements bring the slight improvement or reduction to MultiDRP. A possible explanation for this result is that cosine function and Euclidean Distance have the same ability to measure the similarity distributions among samples. Concurrently, MultiDRP only need to select 10 most similar entities to construct cell lines or drugs.

3.5.3. Selection of neighbors in biological networks

It is interesting and important to verify the effect of the number of neighbors on the performance of MultiDRP. Therefore, we design the experiments which determine the neighbors based on a similarity threshold value, termed MultiDRP-ST. To be specific, the cosine similarity among all cell lines is calculated and ranked. We selected the

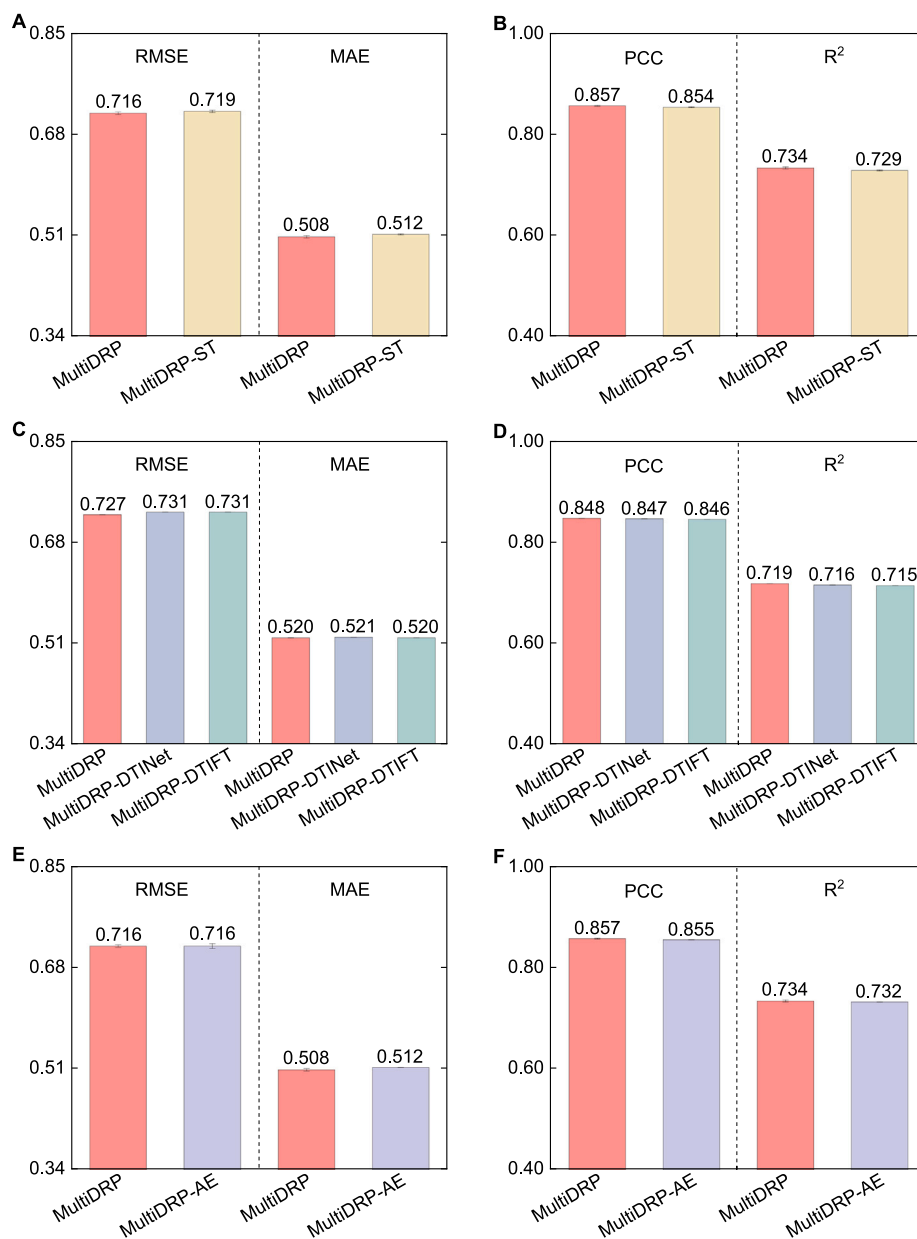


Fig. 7. The impact of key components on MultiDRP results. (A–B) Results based on different methods of neighbor selection. (C–D) Results based on multimodal features. (E–F) Results based on different dimension reduction methods.

similarity score at the top 5% position as threshold value. Therefore, in cell line network, there is an edge between cell lines when its similarity score is higher than the threshold value. Concurrently, we use the same strategies to construct the association network among drugs. As shown in Fig. 7(A–B), we found that the performance of MultiDRP not changed much compared to MultiDRP-ST. Specifically, the difference between MultiDRP and MultiDRP-ST is only 0.3~0.5% in term of various evaluation metrics. It may probably because that both MultiDRP and MultiDRP-ST essentially choose the entities with higher similarity score as neighbors. Therefore, there are the large-scale overlapped edges between biological networks of MultiDRP and MultiDRP-ST. In addition, the biological networks have the small world property [71]. MultiDR can capture high-order relationships by adding more graph attention layers, thus reducing the effect of first-order neighbors on performance.

3.5.4. Selection of multimodal features

It is important to integrate multimodal features including the side effects, and drug–target interactions for drug discovery [72,73].

Therefore, we incorporated drug–target interactions into MultiDRP and proposed MultiDRP-DTIFT and MultiDRP-DTINet (as shown in Supplementary Text S5). As shown in Fig. 7(C–D), we found that the performance of MultiDRP not changed much compared to MultiDRP-DTIFT and MultiDRP-DTINet. In particular, the MAE value of MultiDRP is almost same as the results of MultiDRP-DTIFT and MultiDRP-DTINet. A possible explanation for this result is that the proposed hierarchical attention network proposed by MultiDRP reduce the dependence and sensitivity on the input data to a certain extent.

3.5.5. Selection of dimension reduction methods

To verify the effect of dimension reduction methods on MultiDRP. We use autoencoder to reduce dimension, named MultiDRP-AE, and then is compared with MultiDRP to determine the most suitable dimension reduction methods. As shown in Fig. 7(E–F), we found that MultiDRP-AE generates the similar results to MultiDRP. In other words, there is no significant difference between performance of MultiDRP and

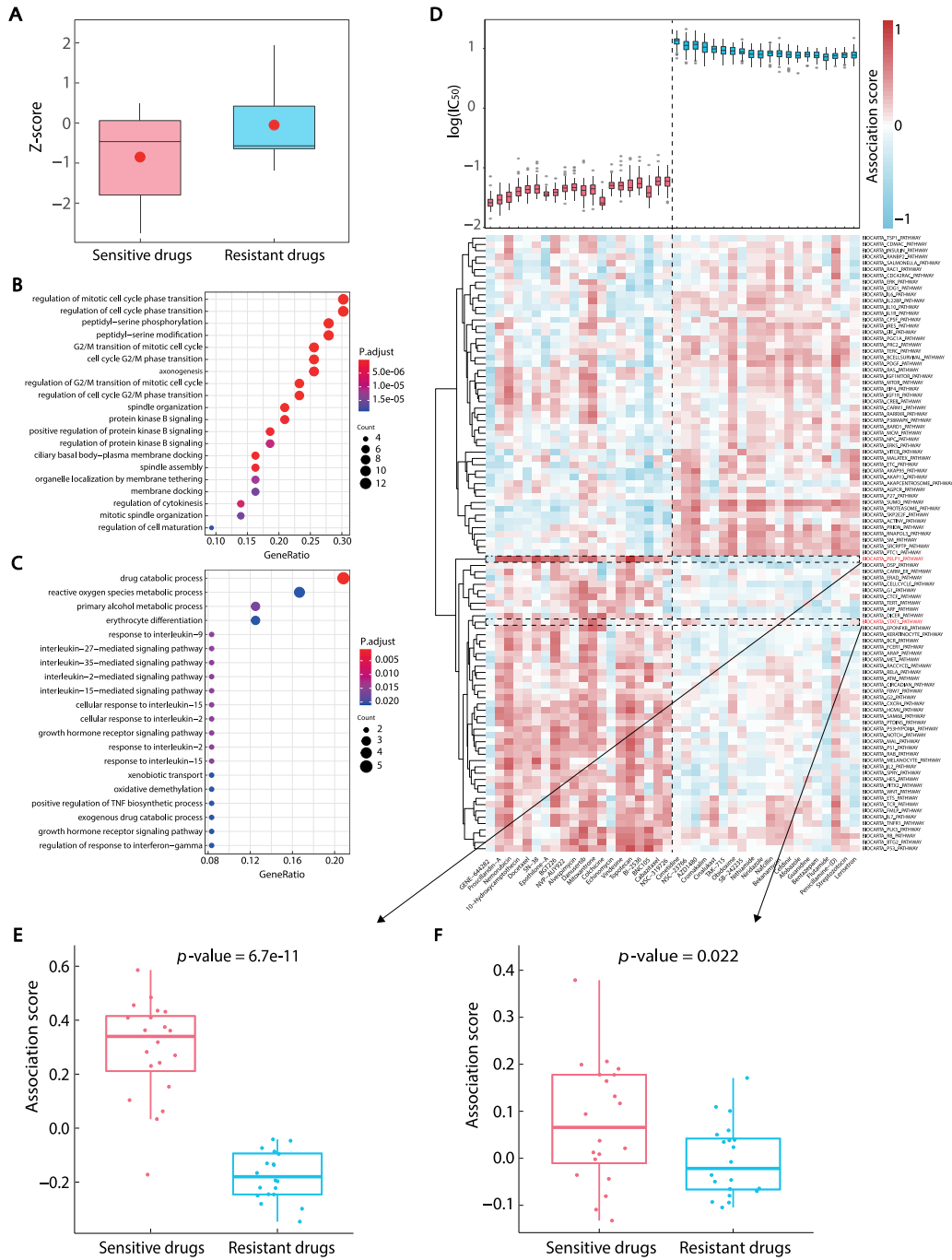


Fig. 8. Comparison of sensitive and resistant drugs. (A) Z-score of sensitive and resistant drugs. (B–C) Gene ontology biological process enrichment of genes of sensitive and resistant drugs, respectively. (D) The heatmap of drug-pathway associations. The boxplot shows the predicted $\log(IC_{50})$ value. (E–F) The association scores of sensitive and resistant drugs between PELP1 and STAT3 pathway, respectively.

MultiDRP-AE. This could probably be attributed to that the similar feature distribution from original data is reserved when using autoencoder and PCA for dimension reduction.

3.6. Great ability distinguishing sensitive and resistant drugs

In addition to the verification based on all known drug-cell line samples, we further analyzed the novel samples predicted by MultiDRP, which are not included in the known cell line-drug response data. The distribution of the newly predicted drug response value ($\log(IC_{50})$) is depicted in Supplementary Figure S7(A). In addition to predicting the exact response value, it is also important to detect

whether cell lines are sensitive or resistant to a given drug [74,75]. Therefore, according to the guidance [76,77], the top 1% cell line-drug pairs that have the smallest $\log(IC_{50})$ were defined as the sensitive group. In contrast, the bottom 1% cell line-drug pairs that has the largest $\log(IC_{50})$ were treated as resistant group. For sensitive group, we visualized the distribution of all drugs on various cell lines and found that there are 10 drugs that are sensitive to more than 150 cell lines, as shown in Supplementary Figure S7(B). These drugs were demonstrated to have great potential for treating a variety of cancers (as described in Table S3), according to data from PubMed literature, ClinicalTrials.gov (<https://clinicaltrials.gov/>), and EU Clinical Trials Register (<https://www.clinicaltrialsregister.eu/>). Among these drugs,

colchicine, which exhibited the repositioning potential, is an interesting example. Although colchicine was approved for the management of gout and exacerbations of Familial Mediterranean Fever, it is being used for the treatment of various cancers, such as lung cancer, gastric cancer, and hypopharyngeal cancer [78,79]. These results suggest that drugs exhibiting high sensitivity to a number of cell lines may play a great role in treatment of multiple cancers.

Compared to agents that are sensitive to cell lines in various tissues, drugs that exhibit high sensitivity in specific tissues are more important for personalized treatments [24]. Simultaneously, we conjectured that if the novel prediction of MultiDRP is accurate, there should be a significant difference between sensitive or resistant drugs. On the other hand, we found that lung tissue includes the most drug-cell line samples, as shown in Figure S3. Therefore, we selected the top 20 drugs which are highly sensitive or resistant to more cell lines in lung tissue, as shown in Supplementary Figure S8. In the following analysis, we compared the effect of the selected sensitivity and resistance drugs on lung tissue by network proximity, GOBP enrichment, and drug pathway association analysis.

Protein–protein interaction network-based proximity has served as a promising approach quantifying the relationship between diseases and drugs [55,57,58]. Therefore, we performed network-based proximity analysis to quantitatively measure the association between the novel predicted drugs and lung cancer (see Methods). To be specific, proteins related to lung cancer tend to form a localized neighborhood in PPI network. Concurrently, for a drug with high sensitivity to lung cancer, its target proteins should be within or in the immediate neighborhoods of the lung cancer proteins in PPI network. Here, network proximity was converted to a Z-score by permutation tests according to the guidance [55,58]. Generally, closer proximity (smaller Z-score) in PPI network represents stronger association between a drug and a disease. As shown in Fig. 8(A), drugs which are sensitive to lung cancer achieved smaller Z-score (closer network proximity) compared with the drugs which are resistant to lung cancer. This phenomenon is also consistent with the distribution of their IC_{50} values predicted by MultiDRP. These results suggest that the sensitive drugs are different from resistant drugs at network proximity perspective.

In addition to comparing the proximity of lung cancer and two kinds of drugs in human PPI, it is also important to summarize the biological processes of targets involved in the sensitive and resistant drugs. Therefore, we carried out GOBP enrichment analysis on the targets of sensitive and resistant drugs, respectively. The enrichment results are shown in Figs. 8(B) and 8(C), where the size of dots corresponds to enriched gene quantity. The color of dots corresponds to the significance of enrichment. For the sensitive drugs, GOBP enrichment analysis revealed multiple biological processes, which are closely correlated with cancer development and metastasis, such as regulation of the cell cycle phase transition and peptidyl-serine phosphorylation. In contrast, for the resistant drugs, we observed that there was no cancer-related biological process. The most significantly enriched biological process was drug catabolic process, which has been demonstrated to be involved in drug resistance in multiple cancers [80,81].

To further explore the different mechanisms of sensitive and resistant drugs, we analyzed the association between drugs and signaling pathways which was defined based on activity scores of pathways and predicted IC_{50} values of drugs (see Methods). For cell lines in lung tissues, IC_{50} value of the sensitive drugs in the left part of Fig. 8(D) is smaller than that of the resistant drugs in the right part. In addition, there was a significant difference in pathways between sensitive drugs and resistant drugs. This suggests that the two kinds of drugs have the different mechanisms. For instance, in terms of the PELP1 pathway, there was a significant difference (p -value = $6.7e-11$) in association scores between sensitive and resistant drugs, as shown in Figs. 8(E) and 8(F). This phenomenon is consistent with previous studies showing that PELP1 pathway could affect drug sensitivity in cancer cells [82]. Similarly, the association scores of the sensitive and resistant groups

were also significantly different (p -value = 0.022) in terms of STAT3. Interestingly, a previous study has made a similar finding that the feedback activation of STAT3 was indeed a cancer drug-resistance mechanism [83]. Naturally, these results further suggest that the sensitive and resistant drugs predicted by MultiDRP exhibit significant differences in biological processes and functions.

3.7. *In vitro* validation of drug response for NCI-H23 cell line

In order to verify the practicability of MultiDRP for specific cell lines, we performed *in vitro* experiment assays based on NCI-H23 which is an important cell line in lung cancer and has been widely used for various lung cancer researches [84,85]. To be specific, MultiDRP screened 10 novel drugs with the highest sensitivity to NCI-H23. Subsequently, we examined the effects of eight drugs on lung cancer cell viability *in vitro* because the other two drugs could not be purchased. The cytotoxic effects of drugs at different concentrations on NCI-H23 cell line are shown in Fig. 9. The IC_{50} values of drugs were calculated by using the dose–response curves. We observed that eight drugs were highly sensitive to the NCI-H23 cell line. In particular, the IC_{50} value between epothilone and NCI-H23 was only 1.66 nM. Among eight candidate drugs, six of these drugs are Experimental or Investigated, and the other two drugs are approved by Food and Drug Administration (FDA). Among the Experimental/Investigated drugs, four drugs (i.e., Epothilone B, Prexasertib, Triptolide, and Luminespib) are being investigated in clinical trials for treatment of lung cancer (Clinical trial registration ID: NCT00171834, NCT02735980, NCT05166616, and NCT01124864). For two approved drugs, their original indications are not related to lung cancer. Mitoxantrone is used for the treatment of secondary progressive, progressive relapsing, or worsening relapsing-remitting multiple sclerosis. Bortezomib is indicated for the treatment of multiple myeloma or mantle cell lymphoma. However, the ClinicalTrials.gov data shows that Mitoxantrone and Bortezomib have been used in trials studying the treatment of lung cancer (ID: NCT04352413, NCT00411593, NCT00068289). Therefore, these two drugs hold promise for application in lung cancer treatment and warrant further clinical investigation. These results further suggest that MultiDRP can be used for predicting drug response of specific cell lines.

To explore the potential mechanism of these drugs, we used their target genes for enrichment analysis. By using Tabula Sapiens, which is a multiple-organ, single-cell transcriptomic atlas of humans, we found that the top enriched terms were the trachea-ciliated cells and lung-ciliated cells [86]. This indicates that these drugs are selective for lung tissue. Furthermore, the GOBP enrichment analysis results showed that the target genes were mainly concentrated in cancer-related biological processes, such as the proteasomal ubiquitin-independent protein catabolic process, cell cycle G2/M phase transition-related process and regulation of cellular amino acid metabolic process. These biological processes have been considered to be important mechanisms for cancer therapy and prevention [87]. In fact, in addition to nemorubicin, other drugs have been used in lung cancer clinical trials according to previous research [88] and ClinicalTrials.gov (<https://clinicaltrials.gov/>). These results indicate that the predicted drugs are sensitive to lung tissue and may play important roles in the future treatment of lung cancer.

3.8. *In vitro* activity of luminespib-pralatrexate combination to lung cancer

The *in vitro* experimental assays suggested that the predicted drugs can affect the viability of the NCI-H23 cell line. However, compared with monotherapy, drug combination has become a more effective and powerful therapeutic in cancer treatment. Therefore, we further identified potential drug combinations for lung cancer. Based on the predicted sensitive drugs for various lung cancer cell lines, multiple drug combinations were screened and verified, which expands the choice of cancer treatment schemes. To improve the efficiency and accuracy of *in vitro* experiments, we first employed a network-based

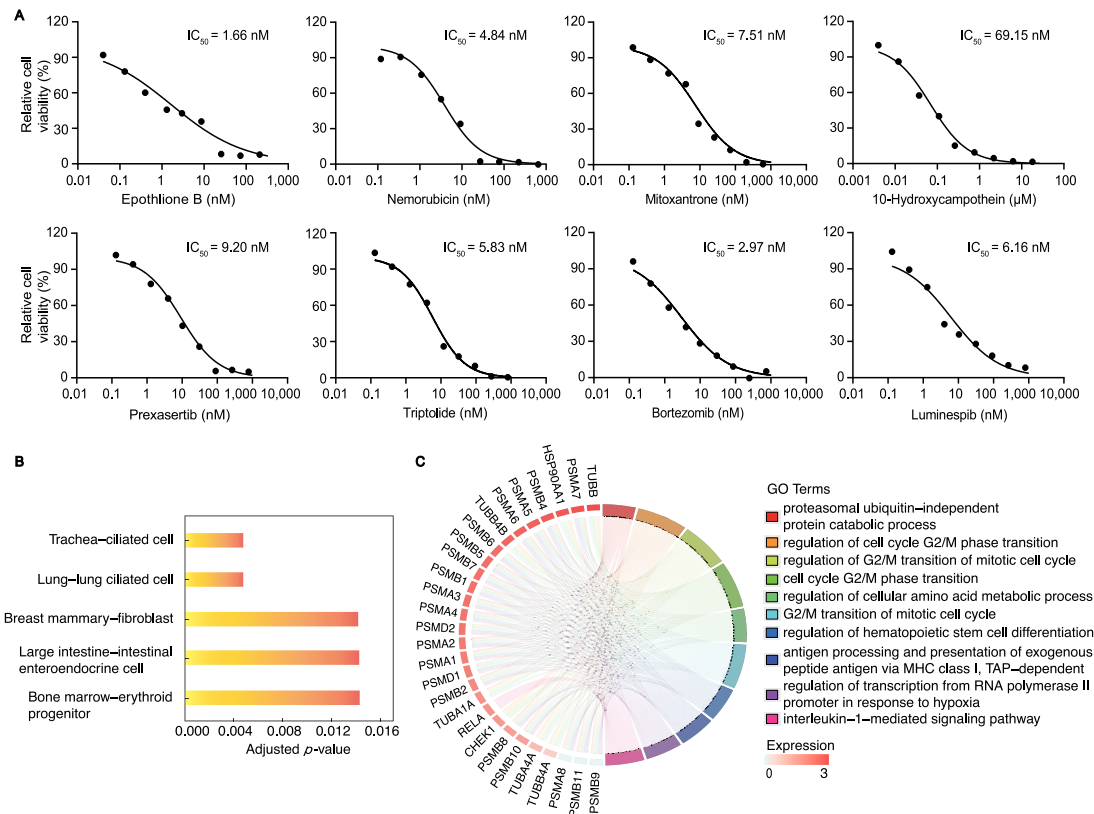


Fig. 9. Validation of novel drug response prediction. (A) Cytotoxic effects of the drugs on the viability of NCI-H23 cells with 5000 cell/well for 72 h of exposure. (B) Enrichment results based on Tabula Sapiens. (C) Gene Ontology Biological Process enrichment results.

methodology, which has been widely used for drug combination prediction of multiple diseases [55,57,58], to reduce the space of drug combinations. To be specific, if the targets of two drugs both hit the disease module, but their target separate neighborhoods, the drug combination could have therapeutically synergistic effects. This pattern was defined as Complementary Exposure (see Methods) [57,58]. Supplementary Table S4 lists the topological association of potential drug combinations with the Complementary Exposure pattern in lung cancer. We selected drug combinations with $Z_{CA} < -1$, $Z_{CB} < -1$ and $S_{AB} \geq 2$ where Z_{CA} and Z_{CB} represent the network proximity between the disease-related gene module and targets of drug A and drug B, respectively, Z_{AB} represents the proximity between modules of drug A and drug B. Among these drug combinations, pralatrexate, whose target gene TYMS has been proposed as an important target in lung cancer [89,90], and HSP90 inhibitors (e.g., luminespib) have been proven to be synergistic with many other drugs in previous studies [91,92]. Thus, we further investigated the combination treatment of luminespib and pralatrexate.

To further explore whether luminespib and pralatrexate have synergistic effects, we carried out a CCK-8 assay to examine the effects of the drug combination on cancer cell viability. Based on the dose-response value obtained by the CCK-8 assay, the Combination Index (CI) of the drug combination was calculated by CompuSyn [93]. The *in vitro* experiments showed that the combination of luminespib and pralatrexate was a promising strategy for the synergistic treatment of lung cancer. The combination of luminespib and pralatrexate showed a strong synergistic effect ($CI < 1$ indicates synergism), as shown in Fig. 10(B). Luminespib has been used for therapeutic of lung cancer [94,95]. Pralatrexate is also being studied for the treatment of lung cancer [96,97]. However, mucositis remains the dose-limiting toxicity of pralatrexate. The individualized dose-modification and prospective mucositis management strategies are key factors to be considered in the future development of pralatrexate. By combining luminespib and

pralatrexate in anticancer treatment, the exposure and tolerability of pralatrexate may be maximized, and the occurrence of mucositis may also be effectively alleviated. Therefore, luminespib combined with pralatrexate is a novel and promising strategy against lung cancers.

4. Discussion

Deep learning is successfully applied to the field of life science [98–102]. In particular, deep learning-based anticancer drug response prediction is important for personalized treatment. Therefore, we propose MultiDRP that develops hierarchical attention networks integrating multi-scale relationships to improve anticancer drug response prediction. MultiDRP simultaneously captures the external relationship among biomedical entities and the internal association among feature items to improve drug response prediction. The comprehensive evaluations on various experiment scenarios demonstrate that MultiDRP outperforms other baseline methods, and achieves the most robust performance. Interestingly, for our datasets, DeepDRK revealed the comparable performance in terms of PCC compared to MultiDRP, but the poor results in terms of RMSE, MAE, and R^2 . These results suggest that DeepDRK is good at predicting the distribution drug-cell line response, but fails to fit the exact drug response values. However, MultiDRP achieved relatively higher performance on all evaluation metrics. On the other hand, for GDSC and TCGA datasets, MultiDRP was superior to DeepDRK on all evaluation metrics. This could probably be attributed to the fact that MultiDRP can capture both the external relationship among entities and the internal correlation of items in cell lines and drugs. In addition, we found that machine learning models tended to achieve lower performance when compared to deep learning-based methods. Nevertheless, the performance of machine learning methods was significantly improved when these models use the representations generated by MultiDRP. The main reason behind this is that the input features from original omics space are full of noise and redundancy,

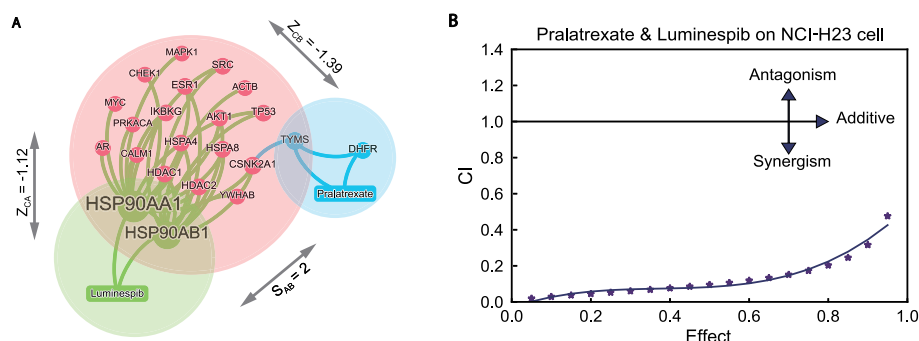


Fig. 10. *In vitro* experimental validation of luminespib and pralatrexate combinations for lung cancer. (A) The inferred mode of action for the combination of luminespib and pralatrexate. The drug combination with the “Complementary Exposure” pattern and may have synergistic effect. (B) Effect-CI plot of the combination of luminespib and pralatrexate on NCI-H23 cell line.

while MultiDRP generates high-quality representations. These results suggest that MultiDRP integrating multi-scale relationships, contributes to improving drug response prediction.

Next, we analyzed the predicted drugs that exhibit high sensitivity or resistance in multiple lung cell lines. The two kinds of drugs exhibited obvious differences in three aspects (i.e., PPI network proximity, GOBP enrichment and drug-pathway association), which reflect their differences in biological mechanisms. This phenomenon is consistent with the fact that the sensitive drugs and resistant drugs play different roles in the treatment of cancer [76,77]. For example, we found that there was a significant difference between sensitive drugs and resistant drugs in the PELP1 pathway. A similar phenomenon was found in previous studies [103] in which PELP1 was overexpressed in lung cancer and promoted cancer cell resistance to tyrosine kinase inhibitor drugs, so silencing PELP1 expression may promote the high sensitivity of lung cancer cells to some drugs. This suggests that the PELP1 pathway may affect drug sensitivity in cancer cells and may be helpful for the development of a new therapy for lung cancer. These results lay a foundation for better understanding of anticancer therapeutics. On the other hand, the difference between the sensitive and resistant group proves the great ability of MultiDRP to distinguish sensitive and resistant drugs. Thus, MultiDRP may be helpful for the development of a new therapy for lung cancer.

Furthermore, through *in vitro* experimental validation on lung cancer, seven out of eight novel drugs exhibited an IC_{50} value of less than 10 nM. The target genes of these drugs were enriched in the trachea-ciliated cells and lung-ciliated cells, which reflects the specificity of these drugs for lung cancer. The GOBP enrichment results also demonstrate that these drugs can be used for the treatment of lung cancer. These results suggest that MultiDRP is an effective tool for drug response prediction. Based on the predicted sensitivity drugs to lung cancer, we found that the combination of luminespib and pralatrexate is a promising strategy for synergistic effect. The drug combination could maximize therapeutic effects and reduce the adverse drug reactions. On other hand, drug sensitivity values predicted by MultiDRP could serve as the important guidance for drug combination discovery. The novel drugs and combinations predicted by MultiDRP should be taken into consideration in clinical test and can be further evaluated as potential therapeutic strategies to lung cancer.

In summary, MultiDRP can achieve great performance in drug response prediction via a multi-scale relationship-driven deep attention network that can integrate both the internal correlation of feature items and the external relationship of biomedical entities. Although MultiDRP is developed to predict the drug response on cell line-level, it can be applied to predict the drug response in patient-level or in other drug discovery studies, such as drug-drug interaction prediction and drug-target interaction prediction. Regretfully, MultiDRP only focuses on cell line networks and drug networks. In other words, MultiDRP

ignores the other associations among biomedical entities, such as, drug-target networks, cancer marker networks or cancer-specific networks, which can naturally simulate the living systems and play a key role in drug development. In the future, we will combine multiple biomedical networks with MultiDRP framework to further improve anticancer drug response prediction. Overall, MultiDRP is a powerful framework to predict drug response and can facilitate experimental validation of sensitive drugs by narrowing the search space for researchers.

5. Conclusion

MultiDRP develops the hierarchical attention networks integrating multi-scale relationship to improve drug response prediction. MultiDRP is appropriate for various prediction scenarios (e.g., large-scale dense data, sparse data, and specific tissue) and achieve great performance by fusing multi-scale relationships. In addition, the sensitive and resistant drugs predicted by MultiDRP exhibit significant differences in biological processes and functions, thus demonstrating that MultiDRP has great discrimination ability. *In vitro* experiments, the sensitive drugs predicted by MultiDRP exhibited high sensitivity to lung cancer cell line NCI-H23. In summary, MultiDRP can serve as a powerful tool for anticancer drug response prediction.

CRedit authorship contribution statement

Xiaoqi Wang: Writing – original draft, Software, Methodology, Conceptualization. **Yuqi Wen:** Writing – review & editing, Methodology, Investigation. **Yixin Zhang:** Validation, Data curation. **Chong Dai:** Validation, Investigation, Data curation. **Yaning Yang:** Software, Investigation. **Xiaochen Bo:** Writing – review & editing, Methodology. **Song He:** Supervision, Methodology, Conceptualization. **Shaoliang Peng:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data needed to evaluate the conclusions in the paper are presented in the paper and the Supplementary Materials. All source datasets and codes can be downloaded from: <https://github.com/pengsl-lab/MultiDRP.git>.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.inffus.2024.102485>.

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